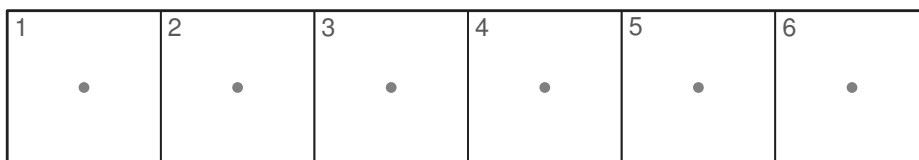
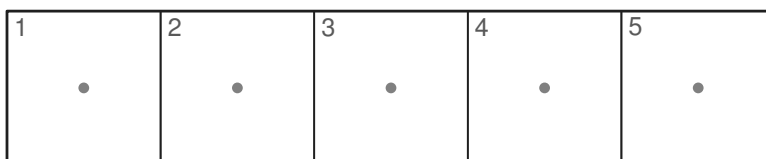
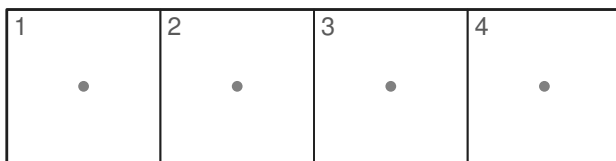


Problem 1: Put a penny on every square of a strip except one; the empty square is the hole. A penny jumps over the penny next to it into the empty square right behind it, and the jumped penny comes off the strip. For each row of the table, set up that hole, jump until no jump is left, and record the fewest pennies you could leave (play again if you think you can do better). When one penny is left, write every square you saw it end on. The table stops at the middle of each strip.



strip	hole on square	fewest pennies left	ending squares seen
four	1		
four	2		
five	1		
five	2		
five	3		
six	1		
six	2		
six	3		

Problem 2: Do the same on the strips of seven and eight squares. Then circle every row of both tables in which one penny was left, and fill in the three lines at the bottom of the page.

1	2	3	4	5	6	7
•	•	•	•	•	•	•

1	2	3	4	5	6	7	8
•	•	•	•	•	•	•	•

strip	hole on square	fewest pennies left	ending squares seen
seven	1		
seven	2		
seven	3		
seven	4		
eight	1		
eight	2		
eight	3		
eight	4		

strip lengths that could be finished with one penny: _____

the holes that worked on them: _____

strip lengths that never could be finished: _____

Problem 3: Colour the squares of this strip: R red, B blue, G green. Set up the hole on square 2. Count the pennies standing on red squares, on blue squares and on green squares and write the three counts in the row marked start. Make one jump and count again. Continue until the game ends, one row per jump, and put the number of pennies on the strip in the last column.

1	2	3	4	5	6
•		•	•	•	•
R	B	G	R	B	G

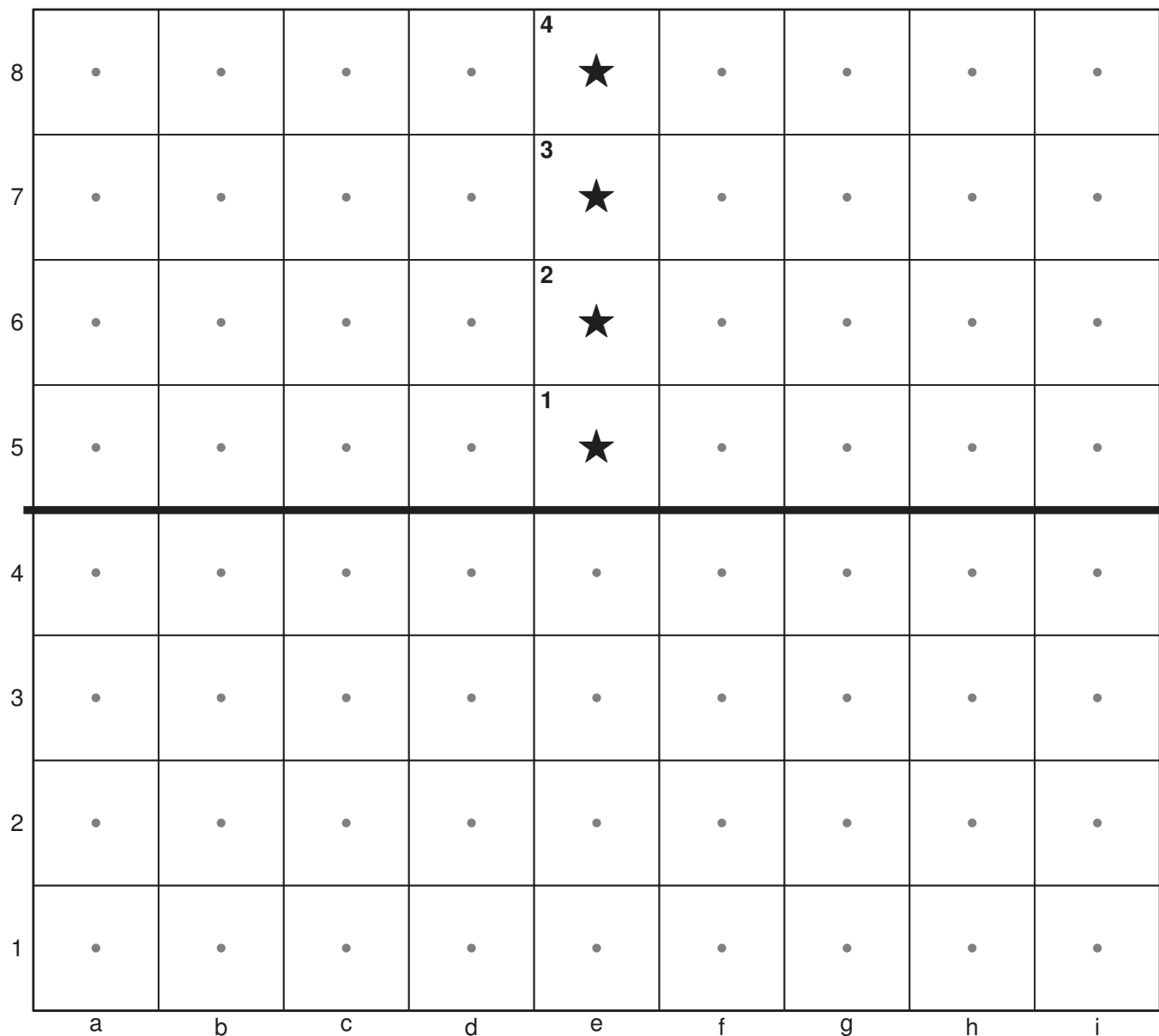
	red	blue	green	total
start				
after jump 1				
after jump 2				
after jump 3				
after jump 4				

Problem 4: Colour the strips of Problems 1 and 2 the same way: squares 1, 4, 7 red, squares 2, 5, 8 blue, squares 3, 6 green. For each row below, set up the hole and write the three counts at the start, whether that hole could be finished (from your tables), and the colour of every ending square you saw.

strip	hole	red	blue	green	finished?	colour of the ending square(s)
four	1					
four	2					
five	1					
five	3					
six	1					
six	2					
six	3					
seven	1					
seven	2					
eight	1					
eight	2					
eight	3					
eight	4					

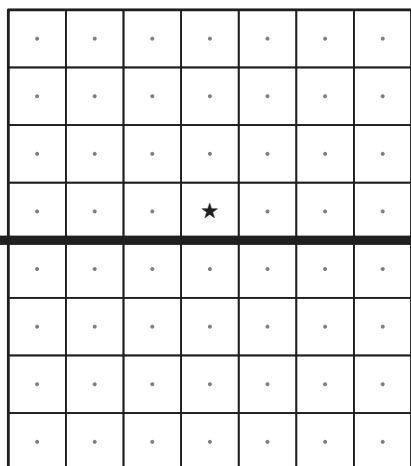
Problem 5: Answer from the tables of Problems 3 and 4. (a) What does one jump do to the red count, to the blue count and to the green count? (b) Write the three counts for a strip with a single penny on a blue square. (c) List every row of Problem 4 in which all three counts are odd, or all three are even. Was any of those holes finished? (d) In the rows that were finished, compare the ending colour with the colour whose count is the odd one out at the start.

Problem 6: The pennies are soldiers. Soldiers start on squares below the thick line, rows 1 to 4, never above it, and jump as before, along a row or a column, never diagonally. Find the smallest army that lands a soldier on star 1, and draw it on the first board of Problem 7. Then find the smallest army for star 2, and then for star 3. Unused soldiers count, so take them off.

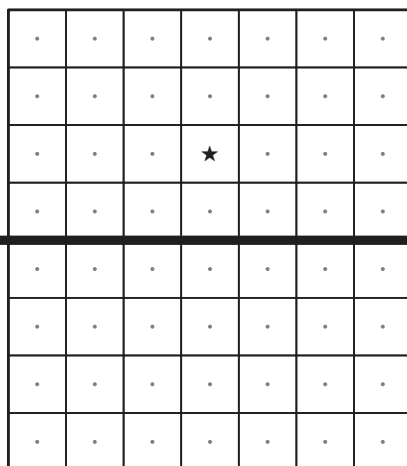


Problem 7: Draw each smallest army you found, a dot for each soldier, and write its size under the board. Write the three sizes in a row and then your guess for the smallest army for star 4.

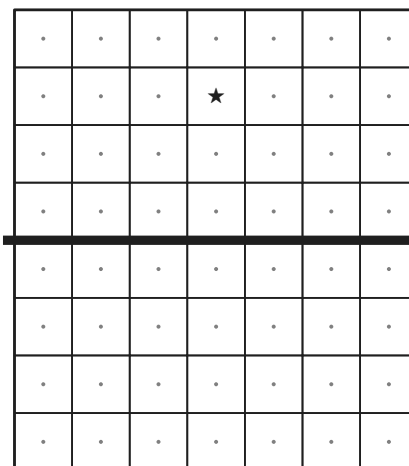
star 1



star 2



star 3



sizes: _____, _____, _____ my guess for star 4: _____

Problem 8: Each square of this strip has a number. (a) Put pennies on the squares numbered 5 and 8 only. Add the numbers under the pennies. Jump the penny on 5 over the 8 onto the 13 and add again. (b) Set up 5 and 8 again, jump the penny on 8 over the 5 onto the 3, and add again. (c) Now put pennies on all six squares left of the thick line and add up the numbers under them. Play the game; after every jump add up the numbers under the pennies. What is the largest total you ever get? (d) Can any army placed on the six squares left of the line get a penny onto the star? Try several armies; for each one, draw a dot on the small strip for each starting penny and write the farthest square a penny reached.

1	1	2	3	5	8	13	21★
---	---	---	---	---	---	----	-----

	total before the jump	total after the jump
(a)		
(b)		

(c) start	after jump 1	after jump 2	after jump 3	the star's number

(d) army at the start

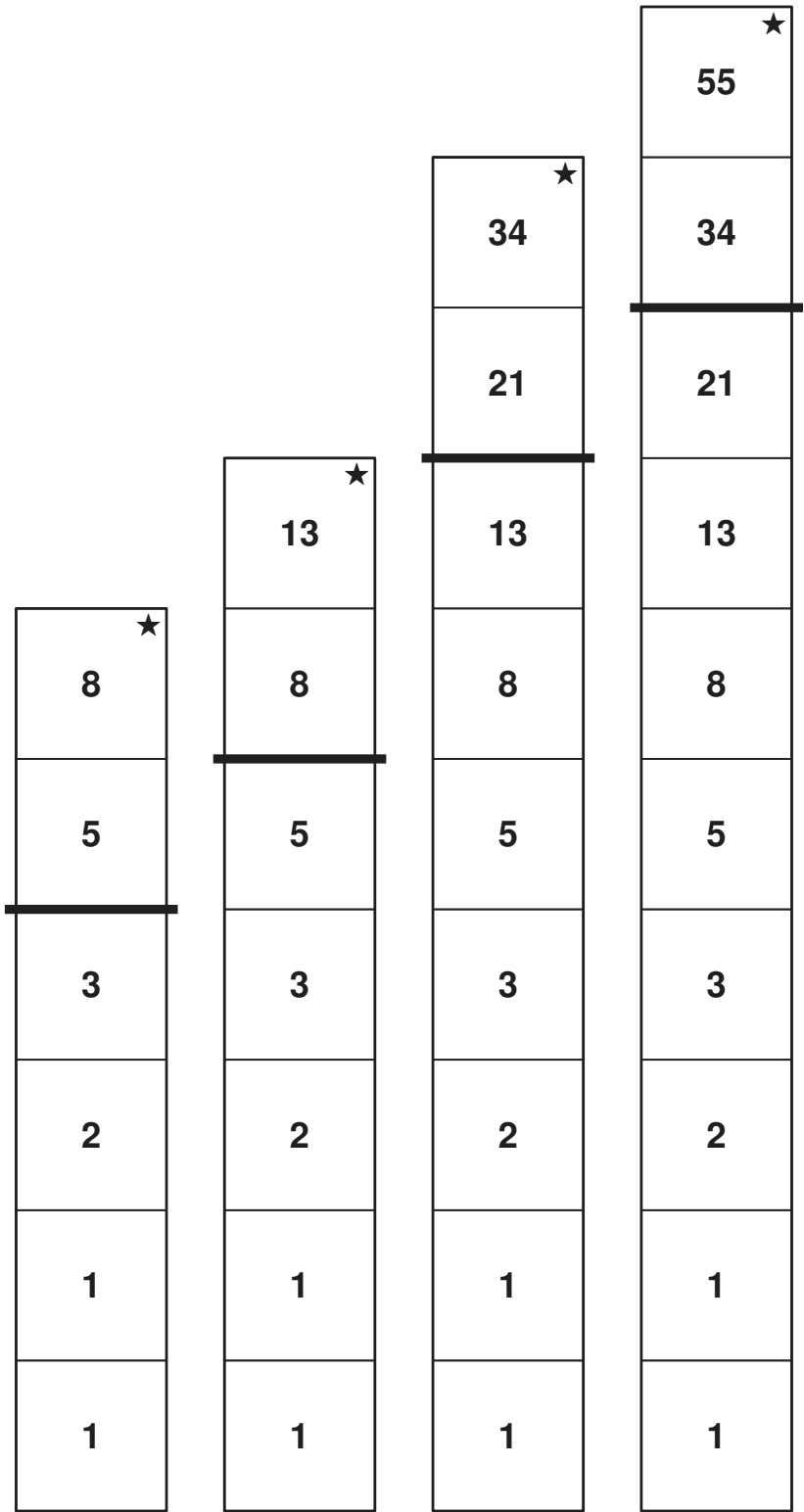
farthest square

								★
--	--	--	--	--	--	--	--	---

								★
--	--	--	--	--	--	--	--	---

								★
--	--	--	--	--	--	--	--	---

Problem 9: Here the army has room for four, five, seven and eight pennies. On each strip, put a penny on every square below the thick line, add up the army’s numbers, and compare with the star’s number. Fill in the table. Then play each strip and write the number of the farthest square past the line that any penny reached.



pennies	army total	star	difference	farthest square
4				
5				
6				
7				
8				

Problem 10: Look at the five differences in your table. Explain, using only what happens to the total at a jump, why no army on these strips can ever reach the star.

Problem 11: Every square of this board carries a weight; the star weighs 1.000. (a) Check these sums to three decimal places: $0.618 + 0.382$, $0.382 + 0.236$, $0.236 + 0.146$, $0.146 + 0.090$. (b) The four shaded squares are your smallest army for star 2. Put pennies on them and add their weights. Make the three jumps that reach the star; after each jump, add the weights under the pennies. (c) Add the three heaviest weights below the thick line. Compare with the star. What does this say about armies of three soldiers?

0.382	0.618	1.000★	0.618	0.382
0.236	0.382	0.618	0.382	0.236
0.146	0.236	0.382	0.236	0.146
0.090	0.146	0.236	0.146	0.090
0.056	0.090	0.146	0.090	0.056

(a)	$0.618 + 0.382$	$0.382 + 0.236$	$0.236 + 0.146$	$0.146 + 0.090$
sum				
(b)	army at the start	after jump 1	after jump 2	after jump 3
weight				

(c) three heaviest below the line: _____

armies of three soldiers: _____

Problem 12: For star 4 the weights are whole numbers that obey the same rule as the strip of Problem 8: along any line toward the star, two neighbouring weights add up to the next one. (a) Check three examples, such as $3 + 5$, $8 + 13$ and $34 + 55$, in different directions. (b) The twenty shaded squares are an army that really does reach star 4. Add its twenty weights and compare with the star. (c) Add the sixteen heaviest weights below the line. What does this say about your guess for star 4? (d) Add the nineteen heaviest weights below the line. Does the weight rule alone settle whether nineteen soldiers can reach star 4?

21	34	55	89	144	★	89	55	34	21
13	21	34	55	89	55	34	21	13	
8	13	21	34	55	34	21	13	8	
5	8	13	21	34	21	13	8	5	
3	5	8	13	21	13	8	5	3	
2	3	5	8	13	8	5	3	2	
1	2	3	5	8	5	3	2	1	
1	1	2	3	5	3	2	1	1	

(b) twenty shaded squares: _____ (c) sixteen heaviest: _____

(c) sixteen soldiers can / cannot reach star 4, because _____

(d) nineteen heaviest: _____ does the rule settle nineteen? _____

Problem 13: The weights for star 3. (a) The shaded squares are one smallest army for star 3. Add its weights. (b) The other smallest army is the five squares of the row just below the line, centred under the star, plus the three squares directly below the left three of them. Add its weights. (c) Add the seven heaviest weights below the line. What does this say about armies of seven soldiers? (d) Put pennies on the shaded army and play it to the star, adding the weights after every jump. Does the total ever change?

0.236	0.382	0.618	1.000 ★	0.618	0.382	0.236
0.146	0.236	0.382	0.618	0.382	0.236	0.146
0.090	0.146	0.236	0.382	0.236	0.146	0.090
0.056	0.090	0.146	0.236	0.146	0.090	0.056
0.034	0.056	0.090	0.146	0.090	0.056	0.034
0.021	0.034	0.056	0.090	0.056	0.034	0.021
0.013	0.021	0.034	0.056	0.034	0.021	0.013

(a) _____ (b) _____ (c) _____

seven soldiers: _____ (d) _____

Problem 14: On the strip of eight with the hole on square 2, which squares does the colouring allow the last penny to end on? Which of them did you actually see in Problem 2? Then predict, from the colours alone, what happens on a strip of ten squares with the hole on square 1, and on a strip of nine with the hole on square 5. Then play them.

1	2	3	4	5	6	7	8	9
.	.	.	.		.	.	.	.

1	2	3	4	5	6	7	8	9	10
	.	.	.	.	.	.	.	.	.

Problem 15: From your table of Problem 2, copy the ending squares for the strip of eight: hole 2 ends on _____, hole 4 ends on _____, hole 5 ends on _____, hole 7 ends on _____ (hole 5 is hole 4 seen from the other end, hole 7 is hole 2). Find every pair of squares such that each one, used as the hole, can end on the other. Write the pairs.

Problem 16: On the strip of eight, play a winning game from the hole on square 2 that ends on square 4, and write its six jumps in the left column ($3>2>1$ means the penny on 3 jumps over 2 onto 1). In the right column copy the same six jumps in reverse order: the last jump first. Now set up the hole on square 4 and play the right column from top to bottom. Is every step a legal jump? Where does the last penny end? Then do the same on the strip of six: a game from hole 2 ending on 5, and its four jumps in reverse order played from hole 5.

1	2	3	4	5	6	7	8
•	•	•	•	•	•	•	•

	my game from hole 2	the same jumps, last first
jump 1		
jump 2		
jump 3		
jump 4		
jump 5		
jump 6		
ends on		

1	2	3	4	5	6
•	•	•	•	•	•

	my game from hole 2	the same jumps, last first
jump 1		
jump 2		
jump 3		
jump 4		
ends on		

Problem 17: The same on a rectangle. A jump goes along a row or a column, never diagonally. Leave square A empty, put pennies on the other eleven squares, and find a game that ends with one penny on square D; write its ten jumps by letter ($C > B > A$). Then set up the hole on D and play the same ten jumps in reverse order. Where does the last penny end?

A •	B •	C •	D •
E •	F •	G •	H •
I •	J •	K •	L •

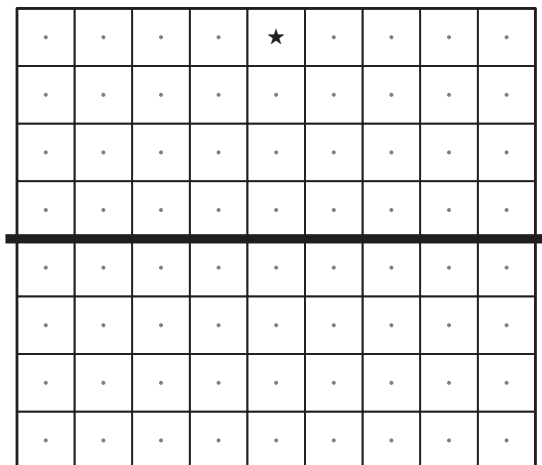
1. _____ 2. _____ 3. _____

4. _____ 5. _____ 6. _____

7. _____ 8. _____ 9. _____

10. _____ reversed, ends on _____

Problem 18: Try to land a soldier on star 4 with an army of sixteen. Draw the army you tried on the board below, write the highest star a soldier reached, and write the fewest soldiers you now think star 4 really needs.



highest star reached: _____

my answer for star 4: _____

Problem 19: For star 2, write the jumps of your smallest army in order, using the coordinates of the board of Problem 6 (e3>e4>e5 means the soldier on e3 jumps over e4 and lands on e5).

Problem 20: Set up the twenty-soldier army on the board of Problem 6: fill row 4 completely (a4 to i4), then d3, e3, f3, g3, then d2, e2, f2, g2, h2, then e1 and f1. Play the nineteen jumps below in order. After jump 16, four soldiers are left, on e5, e6, f6 and g6; watch how the last three jumps use them. Mark each jump as you make it.

- | | | |
|-------------|--------------|--------------|
| 1. e3>e4>e5 | 8. f1>f2>f3 | 15. i4>h4>g4 |
| 2. e1>e2>e3 | 9. c4>d4>e4 | 16. g4>g5>g6 |
| 3. c4>d4>e4 | 10. e3>e4>e5 | 17. e5>e6>e7 |
| 4. a4>b4>c4 | 11. h2>g2>f2 | 18. g6>f6>e6 |
| 5. d2>d3>d4 | 12. f2>f3>f4 | 19. e6>e7>e8 |
| 6. e4>e5>e6 | 13. f4>f5>f6 | |
| 7. f3>f4>f5 | 14. g3>g4>g5 | |

8	.	.	.	.	4 ★	.	.	.	.
7	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.
1	.	.	.	.	.	.	.	.	.
	a	b	c	d	e	f	g	h	i

Problem 21: Every jump in the list brings a soldier two squares closer to the star, counting squares along its row and its column. Using the weights of Problem 12, say why an army that weighs exactly the same as the star can never afford a jump that does not.

Problem 22: A four-by-four board. (a) Leave square B empty, put pennies on the other fifteen, and finish the board with one penny; it takes fourteen jumps. Write the ending square. (b) Now leave square A empty. Copy the digits of pattern 1 onto the big board, one small digit in the corner of each square, and count the pennies standing on squares marked 0, marked 1 and marked 2. What do the counts say about finishing from A? (c) Do the same for the hole on F with pattern 2, and then with pattern 1: does pattern 1 also decide it?

A	B	C	D
E	F	G	H
I	J	K	L
M	N	O	P

pattern 1

2	0	1	2
0	1	2	0
1	2	0	1
2	0	1	2

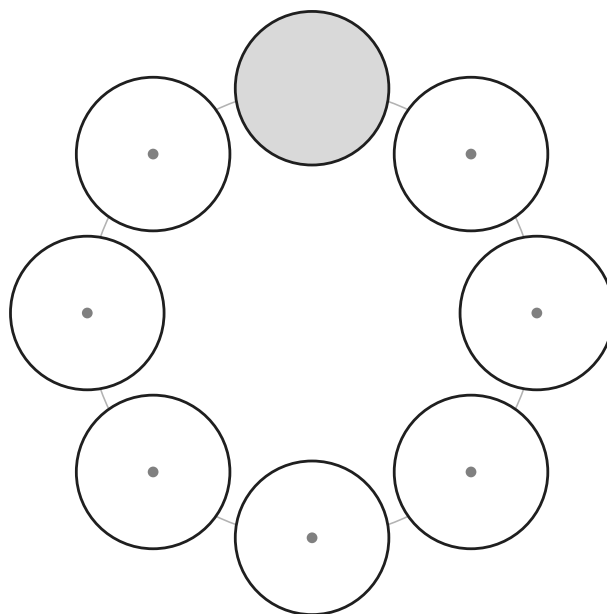
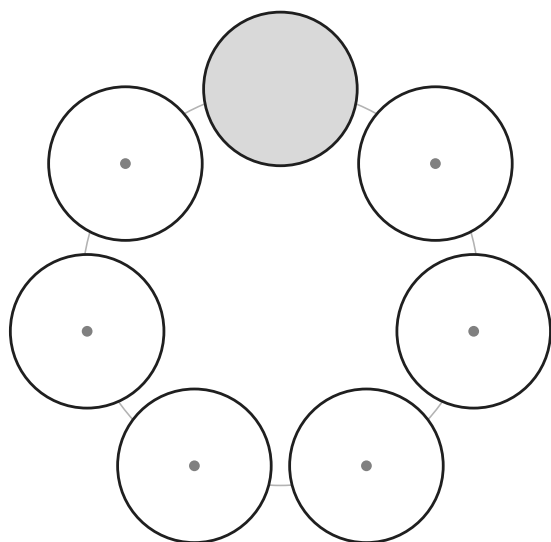
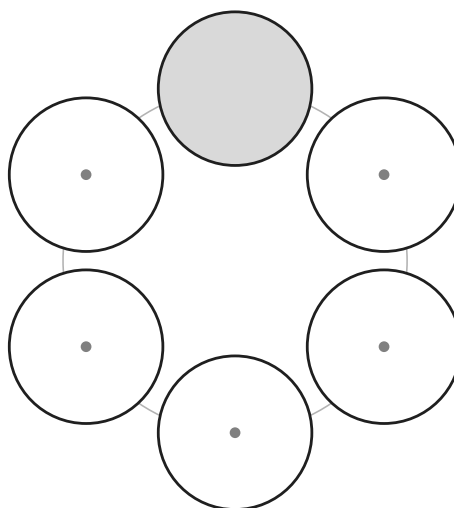
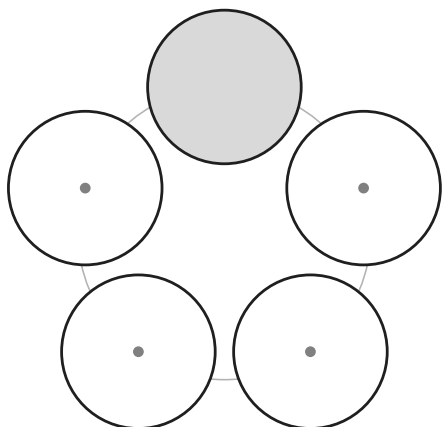
pattern 2

0	2	1	0
1	0	2	1
2	1	0	2
0	2	1	0

hole	pattern	pennies on 0	pennies on 1	pennies on 2	conclusion
A	1				
F	2				
F	1				

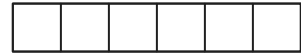
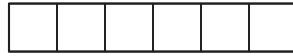
(a) ending square from hole B: _____ Why does a jump on this board change each of the three counts of pattern 1 by exactly one?

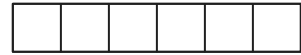
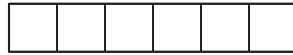
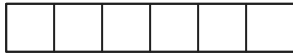
Problem 23: Cells around a ring. Put a penny on every cell except the grey one; a penny jumps over its neighbour into the empty cell beyond, around the ring. For each ring, find the fewest pennies you can leave. Which ring sizes can be finished with one penny?

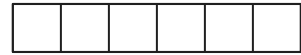
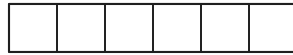
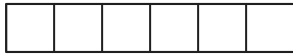


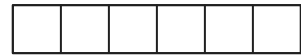
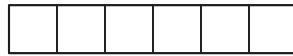
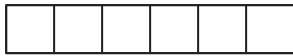
ring of	5	6	7	8
fewest pennies left				

Problem 24: The strip of six with the hole on square 2 can be finished in more than one way. Draw every possible game on the small strips: the first strip shows the start; draw your own arrows from each position to the positions one jump can reach, and write each jump on the line under the strip it leads to. How many winning games are there? Then count the winning games on the strip of eight from the hole on square 2, organising them by their first three jumps.









winning games on the strip of six: _____ on the strip of eight: _____
